

Lab 3

Assignment: SMS Spam Detection Pipeline using PySpark

Objective

In this assignment, you will build a complete **SMS Spam Detection Pipeline** using **PySpark**. The goal is to simulate a real-world data engineering and machine learning workflow by designing an end-to-end pipeline that processes raw text data and produces accurate predictions.

Dataset

Use the **SMS Spam Collection Dataset** (available on Kaggle)

Dataset contains:

- label: spam or ham
- message: SMS text

Project Requirements

Phase 1: Data Ingestion

- Initialize a Spark session
- Load the dataset into a PySpark DataFrame
- Display schema and sample rows

Phase 2: Data Cleaning & Preprocessing

- Rename columns properly
- Handle missing/null values
- Convert labels:
 - spam $\rightarrow$ 1
 - ham $\rightarrow$ 0

Phase 3: Exploratory Data Analysis (EDA)

- Count number of spam vs ham messages
- Analyze message length distribution
- Identify any imbalance in the dataset

Phase 4: Feature Engineering

Use PySpark MLlib to:

1. Tokenize text
2. Remove stopwords
3. Convert text into numerical features using:
 - TF (Term Frequency)
 - IDF (Inverse Document Frequency)

Phase 5: Model Training

- Train at least **one classification model**:
 - Logistic Regression (required)

Phase 6: Pipeline Construction

- Combine all steps into a **single PySpark Pipeline**
- Ensure pipeline is reusable

Phase 7: Model Evaluation

- Split dataset into training and testing sets
- Evaluate using:
 - Accuracy
 - Precision
 - Recall
 - F1-score

Phase 9: Model Saving & Loading

- Save the trained pipeline model
- Reload the model and test it

Phase 10: Prediction on New Data

- Test the model using new SMS messages
- Display predictions